

LEZIONE 7

75/09/25

DISEQUAZIONI

$$\frac{x^3 - 1}{x^4 - 1} > 0$$

$$x \neq \pm 1$$

$$x^4 - 1 \neq 0$$

$$\parallel$$

$$(x^2 - 1)(x^2 + 1)$$

$$x^2 - 1 = 0 \Leftrightarrow x = 1$$

oppo

$$x = -1$$

DIFA. QUADRATI

$$a^2 - b^2 = (a+b)(a-b)$$

DIFA. CUBI

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

NON POSSIAMO

$$D (x^2 - 1)(x^2 + 1)$$

$$\begin{array}{c} + \quad - \quad + \\ | \quad | \quad | \\ -1 \quad 1 \end{array} D$$

DIFA. CUBI

$$N x^3 - 1 \Rightarrow x^3 - 1 = (x - 1)(x^2 + x + 1)$$

$$\begin{array}{c} - \quad + \quad + \\ | \quad | \quad | \end{array} N$$

$$\begin{array}{c} - \quad + \quad + \\ | \quad | \quad | \end{array}$$

Q3 2

$$\frac{2}{x} \geq \frac{1}{x^2 + 1} \quad \boxed{x \neq 0}$$

$$\frac{x^2 + 1}{x} \geq 1$$

NON SI PUO' MOLTIPLICARE PER x (SE POSSO NEGATIVO)
CAMBIA SEGNO

$$\frac{x^2 + 1}{x} - 1 \geq 0$$

$$\frac{x^2 + 1 - x}{x} \geq 0$$

$$D \begin{array}{c} - \quad + \\ | \quad | \\ 0 \end{array}$$

SEMPRE +

$$\begin{array}{c} - \quad + \\ | \quad | \end{array}$$

$$\boxed{S: x > 0}$$

$$x^2 + 1 \leq 2x$$

$$x^2 - 2x + 1 \leq 0$$

$$(x-2)^2 \leq 0$$

$$\boxed{x \geq y}$$

DUADKATU IN BOMPO

$$(a+b)^2 = a^2 + 2ab + b^2$$

GRAFICO

FINDING PEARCE IN VARIABLE FRAMES

$$f: \underset{\text{DOM} \subseteq \mathbb{D}}{0 \in \mathbb{R}} \longrightarrow \underset{\text{CODOM} \subseteq \mathbb{D}}{\mathbb{R}}$$

$$G_0(f) = \{(x, f(x)) \in \mathbb{R}^2 : x \in D\}$$

GRATED ON



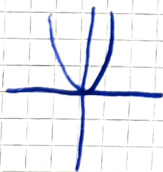
Q: QUANDO UNA CURVA DEL PIANO È UNA FUNZIONE?

PBR 00-1 X UNA SOLA U

OGMI RB $\downarrow$ A VARSILACE

INCONTRA IL GRAFICO AL
PIÙ UNA VOLTA

Ogni RETTA VERTICALE LA
PARSE DAL MINIMO INGIORNA
IL GRAFICO ESAMINANDO
LA CURVA



$$f(x) = x^2$$

NO 12/25

MO 1/12/24



$$f(x) = x^3$$

Bild 2-1

INVERSA

$$\sqrt[3]{x}$$



$\rightarrow$ $\Rightarrow \neg \rightarrow \text{DNA}$



N.



THOUGHTS

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Q 5 E A

IMMAGINE =

PRODOTTO
IN GRADU
TUTTI ADDI 4

CONTR'IMMAGINE

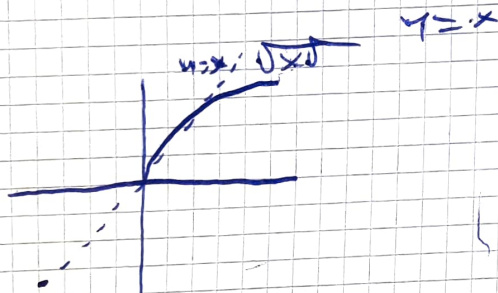
PRODOTTO
= IL SUO INVERSO
X

INIEZIONE $\rightarrow$ AD OGNI y CORRISPONDE
UNA SOLA x

OGNI PENA ORIZZONTALE
INCONTRA IL GRADU AL +
UNA VOLTA

SURIEZIONE $\rightarrow$ AD OGNI y CORRISPONDE
ALMENO UNA x

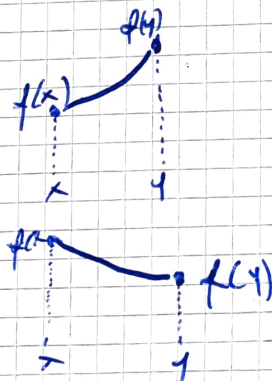
GRADU INVERSA = SIMMETRICO RISPETTO
ALLA BISSETTRICE $y=x$ $\rightarrow$ PER CURVA DISCUTA
RICAVARE LA y



$$y = \frac{1}{x} \quad x \cdot y = 1 \quad x = \frac{1}{y}$$

FUNZIONE CRESCENTE / DECRESCENTE

STRETTAMENTE $\rightarrow$ (NON AMMETTE VALORE)
CRESCENTE SE $f(x) < f(y)$
DECRESCENTE SE $f(x) > f(y)$



STRETTAMENTE
MONOTONA

STRETTAMENTE
MONOTONA
SU
INTERVALLO $\Rightarrow$ INIEZIONE

MONOTONIA

- SE SE MIKE UNA BISSA, DIMONSTRARE
- SE PAVO BASSA FARE 1 ESEMPIO

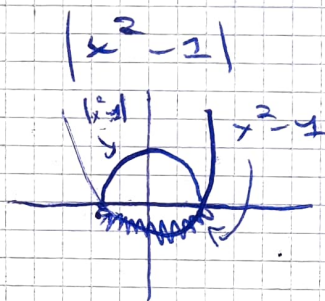
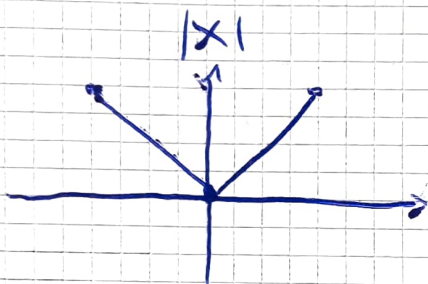
ESEMPIO DI FUNZIONE INiettIVA SU INTERVALLO STRETTAMENTE MONOTONA

questo esempio serve a
"verificare" la affermazione
inversa

f str. monotona su intervallo $\Rightarrow f$ iniettiva

THIS $f \xRightarrow{\text{iniettiva}} f$ STRETTAMENTE MONOTONA SU INTERVALLO

GRAFICO VALORE ASSOLUTO



PERIODICA DI PERIODO T

$$f: D \subseteq \mathbb{R} \rightarrow \mathbb{R}$$

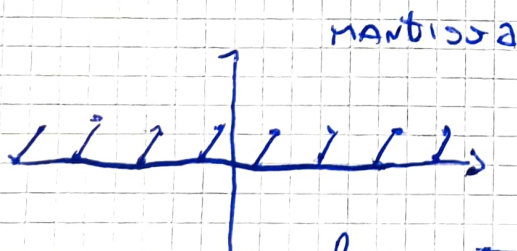
①/2/3/4/5/6/7/8/9/10/11/12/13/14/15/16/17/18/19/20/21/22/23/24/25/26/27/28/29/30/31/32/33/34/35/36/37/38/39/40/41/42/43/44/45/46/47/48/49/50/51/52/53/54/55/56/57/58/59/60/61/62/63/64/65/66/67/68/69/70/71/72/73/74/75/76/77/78/79/80/81/82/83/84/85/86/87/88/89/90/91/92/93/94/95/96/97/98/99/100/101/102/103/104/105/106/107/108/109/110/111/112/113/114/115/116/117/118/119/120/121/122/123/124/125/126/127/128/129/130/131/132/133/134/135/136/137/138/139/140/141/142/143/144/145/146/147/148/149/150/151/152/153/154/155/156/157/158/159/160/161/162/163/164/165/166/167/168/169/170/171/172/173/174/175/176/177/178/179/180/181/182/183/184/185/186/187/188/189/190/191/192/193/194/195/196/197/198/199/200/201/202/203/204/205/206/207/208/209/210/211/212/213/214/215/216/217/218/219/220/221/222/223/224/225/226/227/228/229/230/231/232/233/234/235/236/237/238/239/240/241/242/243/244/245/246/247/248/249/250/251/252/253/254/255/256/257/258/259/260/261/262/263/264/265/266/267/268/269/270/271/272/273/274/275/276/277/278/279/280/281/282/283/284/285/286/287/288/289/290/291/292/293/294/295/296/297/298/299/300/301/302/303/304/305/306/307/308/309/310/311/312/313/314/315/316/317/318/319/320/321/322/323/324/325/326/327/328/329/330/331/332/333/334/335/336/337/338/339/340/341/342/343/344/345/346/347/348/349/350/351/352/353/354/355/356/357/358/359/360/361/362/363/364/365/366/367/368/369/370/371/372/373/374/375/376/377/378/379/380/381/382/383/384/385/386/387/388/389/390/391/392/393/394/395/396/397/398/399/400/401/402/403/404/405/406/407/408/409/410/411/412/413/414/415/416/417/418/419/420/421/422/423/424/425/426/427/428/429/430/431/432/433/434/435/436/437/438/439/440/441/442/443/444/445/446/447/448/449/450/451/452/453/454/455/456/457/458/459/460/461/462/463/464/465/466/467/468/469/470/471/472/473/474/475/476/477/478/479/480/481/482/483/484/485/486/487/488/489/490/491/492/493/494/495/496/497/498/499/500/501/502/503/504/505/506/507/508/509/510/511/512/513/514/515/516/517/518/519/520/521/522/523/524/525/526/527/528/529/530/531/532/533/534/535/536/537/538/539/540/541/542/543/544/545/546/547/548/549/550/551/552/553/554/555/556/557/558/559/560/561/562/563/564/565/566/567/568/569/570/571/572/573/574/575/576/577/578/579/580/581/582/583/584/585/586/587/588/589/590/591/592/593/594/595/596/597/598/599/600/601/602/603/604/605/606/607/608/609/610/611/612/613/614/615/616/617/618/619/620/621/622/623/624/625/626/627/628/629/630/631/632/633/634/635/636/637/638/639/640/641/642/643/644/645/646/647/648/649/650/651/652/653/654/655/656/657/658/659/660/661/662/663/664/665/666/667/668/669/670/671/672/673/674/675/676/677/678/679/680/681/682/683/684/685/686/687/688/689/690/691/692/693/694/695/696/697/698/699/700/701/702/703/704/705/706/707/708/709/710/711/712/713/714/715/716/717/718/719/720/721/722/723/724/725/726/727/728/729/730/731/732/733/734/735/736/737/738/739/740/741/742/743/744/745/746/747/748/749/750/751/752/753/754/755/756/757/758/759/760/761/762/763/764/765/766/767/768/769/770/771/772/773/774/775/776/777/778/779/780/781/782/783/784/785/786/787/788/789/790/791/792/793/794/795/796/797/798/799/800/801/802/803/804/805/806/807/808/809/810/811/812/813/814/815/816/817/818/819/820/821/822/823/824/825/826/827/828/829/830/831/832/833/834/835/836/837/838/839/840/841/842/843/844/845/846/847/848/849/850/851/852/853/854/855/856/857/858/859/860/861/862/863/864/865/866/867/868/869/870/871/872/873/874/875/876/877/878/879/880/881/882/883/884/885/886/887/888/889/890/891/892/893/894/895/896/897/898/899/900/901/902/903/904/905/906/907/908/909/910/911/912/913/914/915/916/917/918/919/920/921/922/923/924/925/926/927/928/929/930/931/932/933/934/935/936/937/938/939/940/941/942/943/944/945/946/947/948/949/950/951/952/953/954/955/956/957/958/959/960/961/962/963/964/965/966/967/968/969/970/971/972/973/974/975/976/977/978/979/980/981/982/983/984/985/986/987/988/989/990/991/992/993/994/995/996/997/998/999/1000

Periodica di periodo T

① t.c. $x+T \in D$ se $x \in D$

$$\textcircled{2} f(x+T) = f(x) \quad \forall x \in D$$

FUNZIONI TRIGONOMETRICHE



f PERIODICA

$\Rightarrow f$ non iniettiva

esempio di f non iniettiva non periodica

$$y = x^2$$

LEZIONE 8

POTENZE

$$x^3 - 4 > 0 \Rightarrow x^3 > 4$$

$$\sqrt[3]{x}$$

$$\Rightarrow x > 4$$

SOMMA PERCHÉ 2⁴ È DISPARI

$$n \in \mathbb{N}$$

$$n > 0$$

$$x^n = x \cdot x \cdot x$$

potenza di esponente e base x

PROPRIETÀ

$$x^n \geq 0 \text{ se } n \text{ è pari}$$

$$(-x)^n = \begin{cases} x^n & \text{se } n \text{ è pari} \\ -x^n & \text{se } n \text{ è dispari} \end{cases}$$

$$-x^n \text{ se } n \text{ è dispari}$$

$$x = x^1 \cdot x^n$$

$$(x^n)^m = x^{nm} = (x^m)^n$$

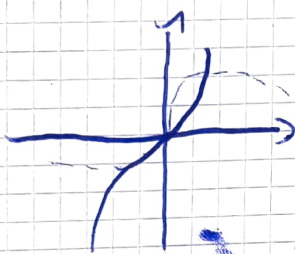
$$(xy)^n = x^n \cdot y^n$$

$$\forall x \neq 0$$

$$x^0 = 1$$

$$x^{-n} = \frac{1}{x^n}$$

N DISPARI



DOMINIO $\mathbb{R}$

IMMAGINE $\mathbb{R}$

STRETT. CRESC.

BISETTIVA

N PARI



DOMINIO $\mathbb{R}$

IMMAGINE $[0, +\infty)$

es

$$n \in \mathbb{N}$$

$$n > 0$$

$$x^n = \frac{1}{x^n}$$

DOMINIO $x \neq 0$

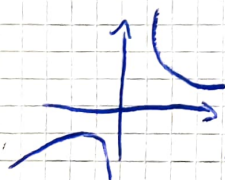
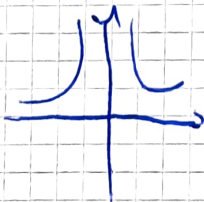


IMMAGINE $\mathbb{R} - \{0\}$



PARI

IMMAGINE $(0, +\infty)$

$$n \in \mathbb{N}$$

$$n > 0$$

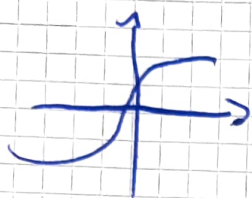
$$x^{\frac{1}{n}} = \sqrt[n]{x}$$

se n è pari

DOMINIO $= [0, +\infty)$

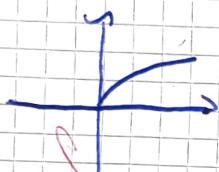
IMMAGINE $= [0, +\infty)$

INVERSA DI x^n



$$\sqrt[n]{x}$$

se n è dispari



se n è pari

si prende solo + se è pari

es 2

$$x^{\frac{m}{n}} = \sqrt[n]{x^m}$$

CRESCe SE $\frac{m}{n} > 0$

DECRESCe SE $\frac{m}{n} < 0$

$$x \neq 0$$

DOMINIO

$(0, +\infty)$

PER OGNI NUMERO REALE SI PUÒ PARE QUALCUNA POTENZA

$\forall x \in \mathbb{R}$

$x > 0$

SI PUÒ PARE QUALCUNA POTENZA

es 3

$$\frac{\sqrt[5]{x^3} = \sqrt[5]{x^3}}{\sqrt[5]{x}} = \frac{\sqrt[5]{x^3}}{\sqrt[5]{x}}$$

$$= \frac{(x^{\frac{3}{5}} \cdot x^{\frac{1}{5}})}{x^{\frac{1}{5}}}$$

$$\boxed{x > 0}$$

$$= \left(x^{\frac{3}{5} + \frac{1}{5}} \right)^{\frac{1}{5}} \cdot x^{-\frac{1}{5}}$$

$$= \left(x^{\frac{3+1}{5}} \right)^{\frac{1}{5}} \cdot x^{-\frac{1}{5}} = x^{\frac{1}{3}} \cdot x^{-\frac{1}{5}}$$

$\sqrt[n]{a(x)} = b(x)$ n DISPARI $\Leftrightarrow$ SIAMO $a(x) = b(x)^n$

n PARI $\Rightarrow \begin{cases} b(x) \geq 0 \\ \sqrt[n]{a(x)} = b(x) \end{cases}$



es 2

$\sqrt{x+2} = x$

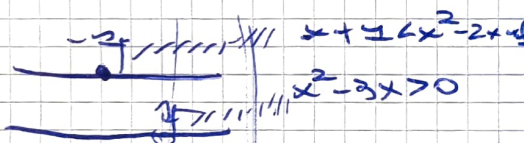
$\begin{cases} x \geq 0 \\ x+2 = x^2 \end{cases}$

$\frac{1 + \sqrt{1+2}}{2} = \boxed{2}$
 ~~$\frac{1 + \sqrt{1+2}}{2}$~~ $\rightarrow$ NON PUO' ESSERE

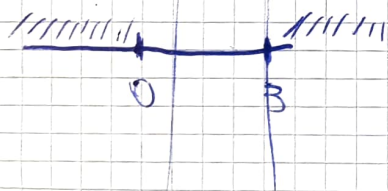
$\sqrt[n]{a(x)} \leq b(x) \Leftrightarrow \begin{cases} a(x) \geq 0 \\ b(x) \geq 0 \\ a(x) \leq b(x)^n \end{cases}$

es 3

$x+1 < x-1$



$\begin{cases} x+1 > 0 \Rightarrow x > -1 \\ x-1 > 0 \Rightarrow x > 1 \\ (x+1)^2 < (x-1)^2 \end{cases}$

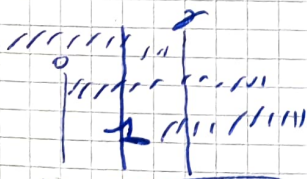


$\boxed{x > 3}$

es 4

$x > \sqrt{2-x}$

$\begin{cases} 2-x \geq 0 \\ x \geq 0 \\ x^2 > 2-x \end{cases}$



$\boxed{1 < x \leq 2}$

$\Rightarrow x^2 + x - 2 = 0$

$\sqrt{x} = 1$

$x < -2$

$x > 1$
 $x < -2$

$$\begin{array}{c}
 \neg \sqrt{a(x)} > b(x) \Leftrightarrow \begin{cases} a(x) \geq 0 \\ b(x) < 0 \end{cases} \quad \vee \quad \begin{cases} b(x) \geq 0 \\ a(x) > b(x) \end{cases} \\
 \text{N PADI} \qquad \qquad \qquad \text{OPPOSITE}
 \end{array}$$